Electric Current

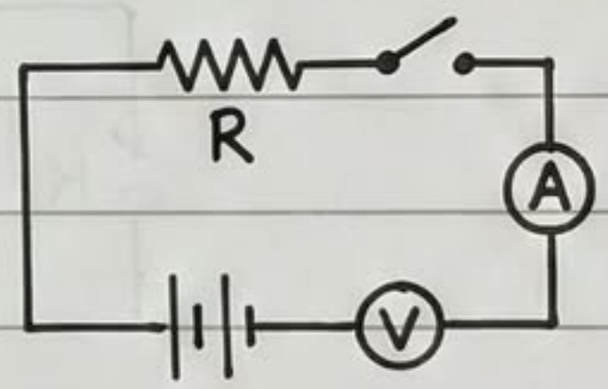
Meaning: Electric current is the flow of electric charge.

Formula: $\boxed{I = \frac{Q}{t}}$

Where I = Current, Q = Charge, t = Time

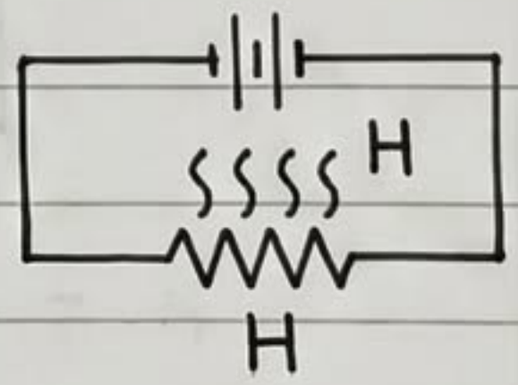
Unit: Ampere (A)
1 Ampere = 1 Coulomb / second

Ohm's Law: $\boxed{V = IR}$



Where V = Voltage,
I = Current,
R = Resistance

Heating Effect of Current:



$$H = I^2 R t$$

(Heat produced depends on square of current, resistance and time)

Ohm's Law

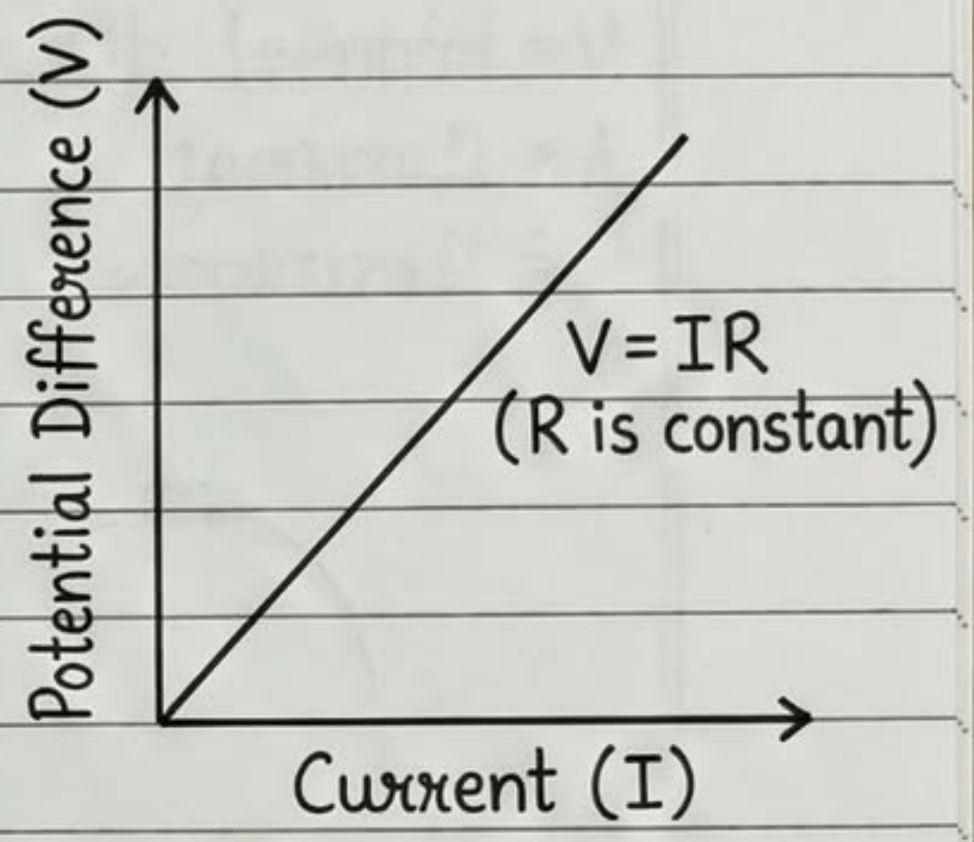
Ohm's Law states that at constant temperature, the current flowing through a conductor is directly proportional to the potential difference applied across its ends. This relationship was given by Georg Simon Ohm.

According to Ohm's Law:

$$V \propto I$$
$$V = IR$$

Where:

V = Potential difference (Volt)
I = Current (Ampere)
R = Resistance (Ohm)



V-I Graph for a Metallic Conductor

This means if the voltage increases, the current also increases provided resistance remains constant.

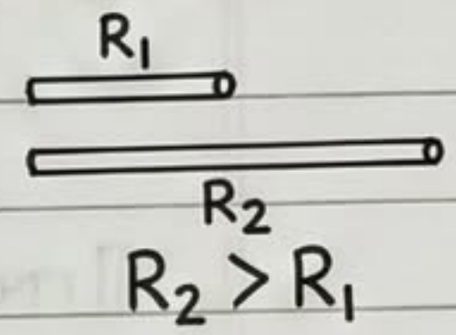
Ohm's Law is valid only when physical conditions like temperature remain constant.

It is widely used to calculate current, voltage and resistance in electric circuits.

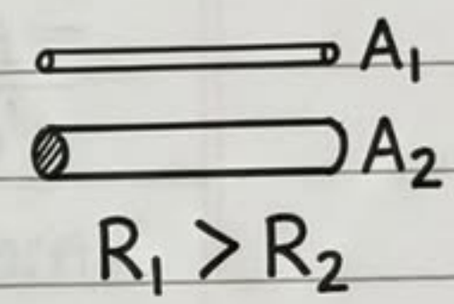
Resistance

Resistance is the property of a conductor that opposes the flow of electric current. It is denoted by R and measured in Ohm (Ω).

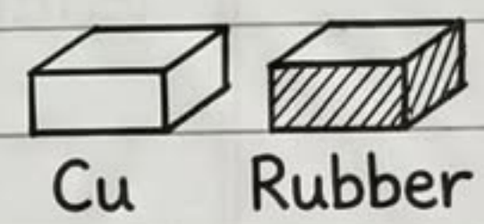
- The resistance of a conductor depends on:
  - Length of the conductor (L)
    Resistance increases with length.



  - Area of cross-section (A)
    Resistance decreases with increase in area.



  - Nature of material
    Different materials have different resistivities (ρ).



  - Temperature (T)
    For metals, resistance increases with rise in temperature.

$$R = \frac{\rho L}{A}$$

Where:
ρ = Resistivity of the material
L = Length
A = Area of cross-section

Materials	Resistance	Examples
Conductors	Very low	Copper, Aluminium
Insulators	Very high	Rubber, Plastic

Electric Power

Electric power is the rate at which electrical energy is consumed or converted into other forms of energy. It is denoted by P and measured in Watt (W).

Formulas:

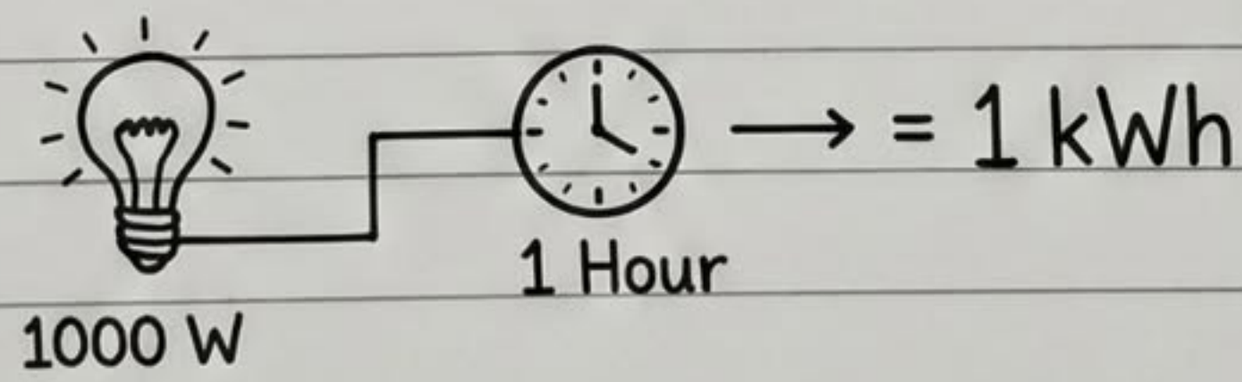
$$P = V \times I$$

$$P = I^2 R \quad \text{(Using } V = IR\text{)}$$

$$P = \frac{V^2}{R} \quad \text{(Using } I = V/R\text{)}$$

One watt is the power consumed when one ampere of current flows through a conductor at a potential difference of one volt.

Commercial unit of electrical energy is kilowatt-hour (kWh).

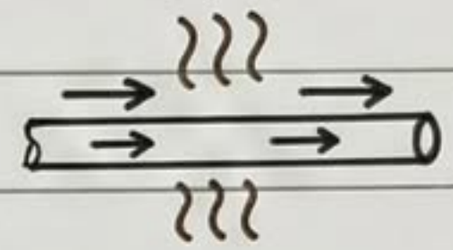


Electric power determines how much electricity an appliance consumes.

Heating Effect of Electric Current

When electric current passes through a conductor, heat is produced. This is known as the heating effect of electric current.

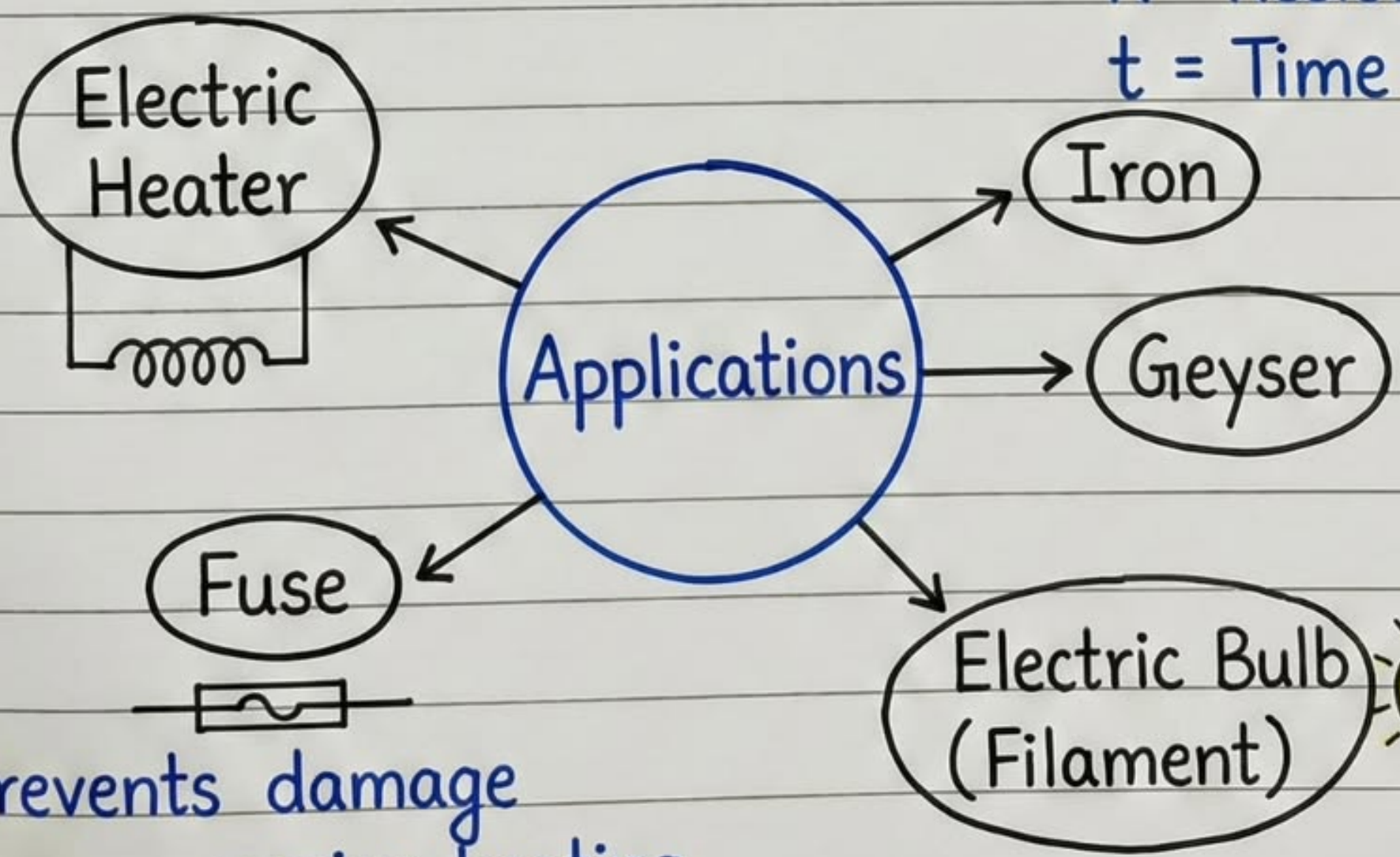
- The heat produced in a conductor depends on:
  - Square of current (I^2)
  - Resistance (R)
  - Time (t)



- According to Joule's Law of Heating:

$$\boxed{H = I^2Rt}$$

Where: H = Heat produced
I = Current
R = Resistance
t = Time

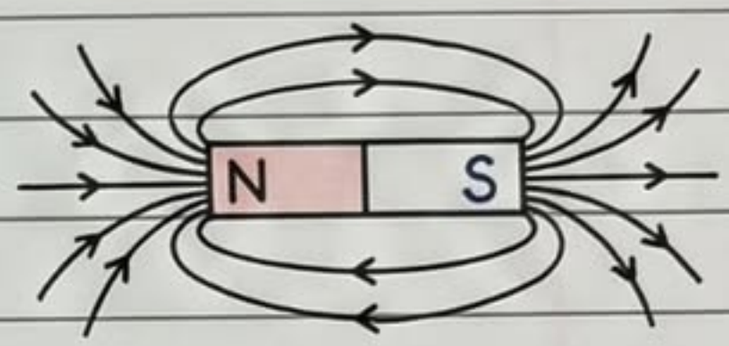


Magnetic Field Lines

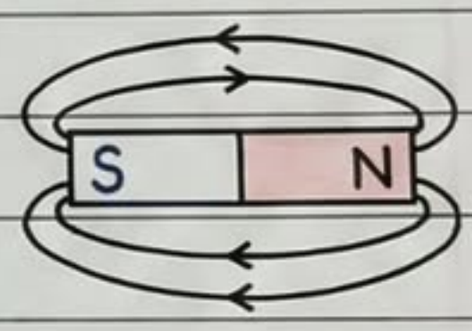
A magnetic field is the region around a magnet where its force can be felt. Magnetic field lines are imaginary lines used to represent the magnetic field.

Properties of magnetic field lines:

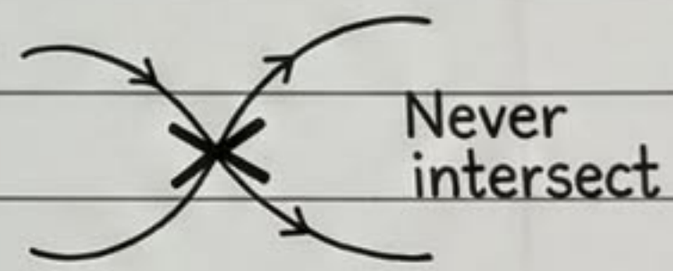
a) They emerge from the North pole and enter the South pole outside the magnet.



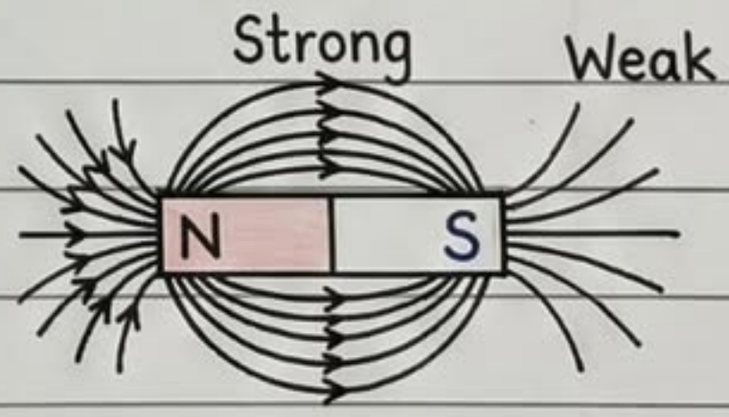
b) Inside the magnet, they move from South to North, forming closed curves.



c) They never intersect each other.



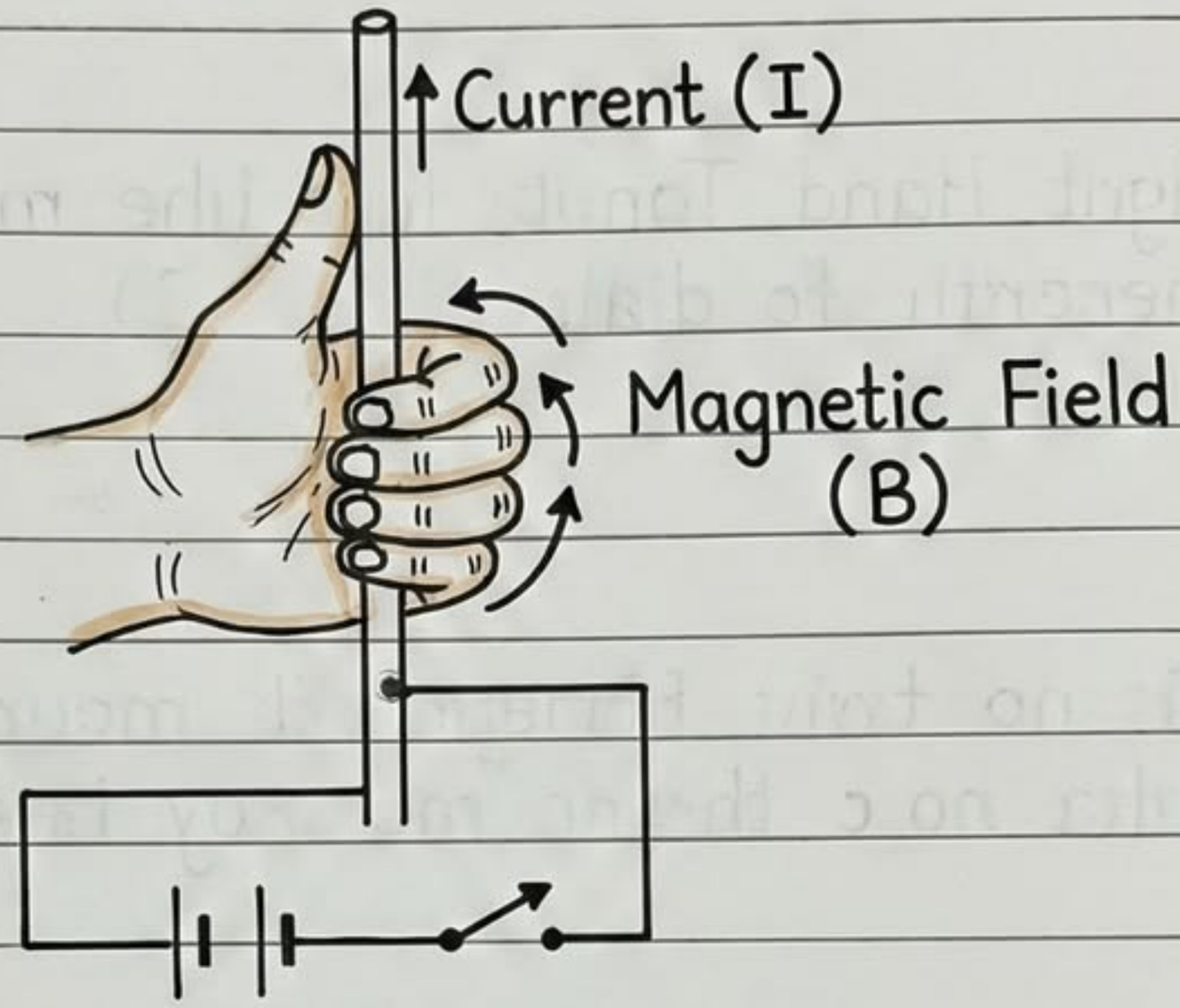
d) The closer the lines, the stronger the magnetic field.



Magnetic field lines help us understand the strength and direction of magnetic force around a magnet or a current-carrying conductor.

Right Hand Thumb Rule

The Right Hand Thumb Rule is used to determine the direction of magnetic field around a straight current-carrying conductor.



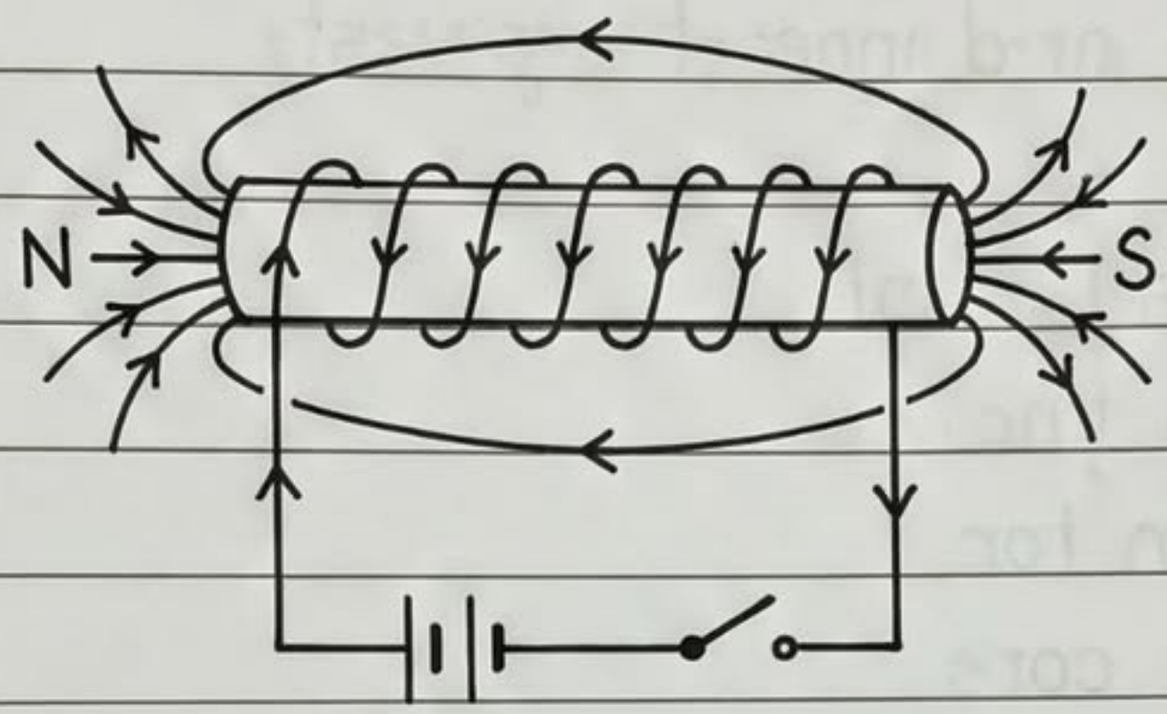
Rule: According to this rule, if we hold the conductor in the right hand such that the thumb points in the direction of current, then the curled fingers represent the direction of magnetic field lines.

Current (I) Direction → Thumb Points → Curled Fingers Show Magnetic Field (B) Direction

This rule helps in understanding the circular magnetic field formed around a current-carrying wire.

Electromagnet

An electromagnet is a temporary magnet produced when electric current flows through a coil of wire wound around a soft iron core.



Strength of an electromagnet depends on:

a) Number of turns in the coil ↑ More turns = Stronger

b) Strength of current ↑ More current = Stronger

c) Nature of core material → Soft iron core = Stronger

Electromagnets are used in electric bells, cranes for lifting heavy iron objects, relays and motors. They are very useful because their strength can be controlled by changing the current.